

IC 221: Foundations of Electrodynamics

Course Code	IC 221
Course Name	Foundations of Electrodynamics
L-T-P-C	2.5-0.5-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum; 47th BoA (Sep 2022); 37th Senate (Oct 2022)
Recoded to	EP 321 by the 60th BoA (Jun 2025)

Course Contents

- **Derivation of Maxwell's equations**

(Review) Vector calculus, Helmholtz equation, Coulomb's law, Gauss law, Poisson and Laplace equations. [3 Lectures]

Electrostatic boundary conditions, Conductors and capacitors, mean value and uniqueness theorem, separation of variables, Dipoles and electric polarization in matter Dielectrics. [6 Lectures]

Lorentz force law – Biot and Savart law and Magnetic vector potential – boundary conditions on B. Magnetic materials – paramagnetic, diamagnetic. Bound currents – boundary conditions on H, Inductance – magnetic energy density [6 Lectures]

Ohm law – EMF's – Faraday's law - Maxwell's equations [5 Lectures]

- **Maxwell's equations and electromagnetic waves**

Electromagnetic waves in vacuum - Maxwell's stress tensor – momentum conservation Poynting theorem and conservation of energy and momentum [5 Lectures]

Gauge transformations, Coulomb gauge and Lorentz gauge. [3 Lectures]

Electromagnetic waves in matter – reflection, transmission, polarization - Electromagnetic waves in dispersive medium – KramersKronig relation - Lorentz oscillator model for atomic dispersion and absorption, negative-index materials [6 Lectures]

Waveguides, transverse electric and transverse magnetic modes, Radiated power, Electric dipole radiation, antenna theory [6 Lectures]

Text Books

1. D J Griffiths, **Introduction to electrodynamics** .

References

1. R P Feynman, **Lectures on Physics II**.
2. D K Cheng, **Fields and wave electromagnetics**.
3. M. O Sadiku, **Elements of Electromagnetics**
4. Purcell E M, **Electricity and Magnetism**.
5. B. B. Laud, **Electromagnetism**.
6. J. D Jackson, **Classical electrodynamics**